Project Assignment: Social Media Application

Introduction

In this project, you will work in **groups of 3 members** to design and develop a **Social Media Application** using **Java**, **MySQL**, **OOP**, **Data Structures & Algorithms**, and **JavaFX GUI**. The application will allow users to create profiles, post updates, interact with others, and perform various social media activities. This project will test your ability to integrate multiple technologies and concepts into a cohesive and functional software system.

Project Requirements

1. Core Features

The application must include the following **core features**:

1.1 User Registration and Login

- **Description**: Users should be able to register and log in to the application.
- **Details**:
  - Store user details (e.g., name, email, password) in a MySQL database.
  - Use OOP concepts to model the `User` class.
  - Implement secure password hashing.
  - Provide a JavaFX GUI for registration and login.

1.2 User Profile Management

- **Description**: Users should be able to create and update their profiles.
- **Details**:
  - Allow users to upload a profile picture, add a bio, and update personal information.
  - Store profile data in the MySQL database.
  - Use OOP concepts to model the `Profile` class.
  - Provide a JavaFX GUI for profile management.

1.3 Posting Updates

- **Description**: Users should be able to post updates (text, images, or both).
- **Details**:
  - Store posts in the MySQL database.
  - Use OOP concepts to model the `Post` class.
  - Provide a JavaFX GUI for creating and viewing posts.
  - Implement data structures like Lists or Arrays to manage posts.

1.4 News Feed

- **Description**: Users should be able to view a news feed of posts from other users (friends).
- **Details**:
  - Fetch posts from the database and display them in a JavaFX GUI.
  - Use pagination or lazy loading for large datasets.

1.5 Likes and Comments

- **Description**: Users should be able to like and comment on posts.
- **Details**:
  - Store likes and comments in the MySQL database.
  - Use OOP concepts to model the `Like` and `Comment` classes.
  - Provide a JavaFX GUI for liking and commenting on posts.

1.6 Friend System

- **Description**: Users should be able to add friends and view their profiles.
- **Details**:
  - Store friend relationships in the MySQL database.
  - Provide a JavaFX GUI for managing friends.

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2. Advanced Features

The application should also include at least **3 advanced features** from the following list:

2.1 Notifications

- **Description**: Users should receive notifications for likes, comments, and friend requests.
- **Details**:
  - Store notifications in the MySQL database.
  - Use OOP concepts to model the `Notification` class.
  - Provide a JavaFX GUI for viewing notifications.

2.2 Search Functionality

- **Description**: Users should be able to search for other users or posts.
- **Details**:
  - Provide a JavaFX GUI for searching.

2.3 Privacy Settings

- **Description**: Users should be able to set privacy levels for their posts (e.g., public, friends-only, private).
- **Details**:
  - Store privacy settings in the MySQL database.
  - Use OOP concepts to model the `Privacy` class.
  - Provide a JavaFX GUI for managing privacy settings.

2.4 Real-Time Chat

- **Description**: Users should be able to chat with their friends in real-time.
- **Details**:
  - Implement a simple chat system using sockets or a library.
  - Store chat history in the MySQL database.
  - Provide a JavaFX GUI for chatting.

Deliverables

1 - Source Code: Github or any similar version control

2 - Jar file to run: Build the JavaFX project as Jar file runnable.

3 - MySql Database: database file (.sql)